Qn. Set Code-7

Semester: 5th
Programme: B.Tech
Branch: CSE, CE, EE,

AUTUMN END SEMESTER EXAMINATION-2025

5th Semester B.Tech

ENGINEERING ECONOMICS

HS30101

(For 2023 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) The demand function for good X is

$Q_x = 2000 - 3P_x + 4P_y$

where,

Q_x = demand for good X

P_x = Price of good X

P_y = Price of good Y

How good X and good Y are related? Give argument to support your answer.

(b) The demand equation for a product is given as P = 50 – 0.5Q (P is price and Q is quantity). Find the P and Q when Marginal Revenue (MR) would be zero.

(c) Write the budged equation and the slope of the budget line using the following notations for the goods X and Y:

P_x = Price of good X

P_y = Price of good Y

M = Money income of the consumer

(d) Write the relationship between Total Product (TP) and Marginal Product (MP) in case of Law of Variable Proportion.

(e) A manufacturing unit uses Labour (L) at the price w and Capital (K) at the price r. At a certain point when the producer equilibrium was checked, it was found that $\frac{MP_L}{w} < \frac{MP_K}{r}$. What alteration should be done now between L and K to reach equilibrium?

(f) Kutz Sarthak deposited a sum of $50000 in his account for a period of 12 years now that pays 8% interest annual compounding. His wife Julia persuaded him to make another deposit of $30000 at the end of 7th year for receiving a bigger maturity amount. If Kutz had to withdraw $80000 from this account at the end of 9th year to buy gold for Julia, what final amount he will get when the account matures?

(g) Find the effective rate of interest of 8% nominal interest compounded semi-annually. How do you interpret this nominal and effective interest rate?

(h) The cost of an asset is $1000000 with a salvage value of $200000 at the end of its life period of 16 years. The fixed depreciation earns interest at 12% compounded annually. Find the book value of the asset at the end of 13th year by Sinking Fund method.

(i) What are the equilibrium conditions for a perfectly competitive firm? Present the equilibrium condition with a proper figure.

(j) How does a rise in the bank rate affect your borrowing from the banks?

SECTION-B

2. (a) It is observed that in India there is consistent rise in the petrol price in recent times because of an increase in global crude oil prices, supply cuts by OPEC, war situations in the Middle East and other geopolitical factors. In the last month the price of petrol increased from ₹100 to ₹120 per litre in the metro cities of the country. As a result of the increase in price, the demand for petrol decreased from 2000000 litres to 1900000 litres per day. Given this position of market demand and price of petrol, reply to the following queries: [5]

(i) Calculate the price elasticity of demand for petrol and indicate whether petrol is less or more elastic in the metro cities.

(ii) Write two important factors influencing the degree of price elasticity in this case. Can you give a suggestion so that the consumers demand will be more elastic for petrol?

(b) Currently Samsung Electronics is the world's largest Smart phone manufacturer. In 2025 the sales of Galaxy S25 Ultra increased by 20% in India. You being an engineering economist observed that this increase in sales happened for a 5% decrease in price of S25 Ultra, a 4% increase in per capita income and 15% increase in the price of its competitive brand Apple iPhone16. Income elasticity for S25 Ultra is +2.0 and cross price elasticity between Galaxy S25 Ultra and Apple iPhone16 is +0.40. [5]

For the above market condition find:

(i) The effect of increase in per capita income and increase in price of Apple iPhone16 on the sales of S25 Ultra.

(ii) Price elasticity value for S25 Ultra. On the basis of the price elasticity value what suggestions you will give to Samsung Electronics if its target is to increase the Total Revenue.

3. (a) The market demand and supply for a particular brand of cigarette mostly consumed by the young mass in India are [5]

$Qd = 1200 - 8P$ (Demand)

$Qs = 200 + 2P$ (Supply)

Since the objective of the government is to reduce the consumption of tobacco products, it decides to impose a 30% indirect tax per pack of cigarette on the manufacturer. If government can't collect more revenue through this tax, it can discourage the consumption of such harmful goods. Considering this step of the government you respond to the following inquiries:

(i) Equilibrium Price and Quantity before and after the tax. Who is taking a higher tax burden, the consumer or the producer?

(ii) Explain the distribution of tax burden with the help of a suitable diagram.

(b) Chunali, a software engineer desires to accumulate a sum of ₹2000000 at the end of 8th year from now for her marriage. Her account is growing at 7.5% interest compounded annually. One of her best friend Junali wants the same amount of money at the end of the same time period for her brother's higher education. Junali's account also grows at the same 7.5% interest annual compounding. As you understand time value of money, help Junali in finding the following: [5]

(i) The single deposit Junali has to make to get this future sum.

(ii) Annual equivalent amount of this single deposit Junali has to deposit to realize the desired sum. Check that Junali receives her desired amount at the end of the deposit period by depositing the annual equivalent amount.

4. (a) KIMS Bhubaneswar, a reputed Hospital in Eastern India is planning to expand its health care facilities by establishing a new 500 bedded critical care unit. The details of such a specialized unit are given below: [5]

Initial Investment – ₹ 100 00 00 000

Life of the project (First phase) – 20 years

Cash inflows – ₹ 10 00 00 000 each year for first 10 years and ₹ 12 00 00 000 each year for the next 10 years

Salvage value – ₹ 20 00 00 000

Calculate the Net Present Value (NPV) of this investment proposal for KIMS and suggest whether KIMS should go ahead with the proposal. Use a Minimum Attractive Rate of Return (MARR) 12% compounded per annum.

(b) A company has to decide about the purchase of a new machine to increase the efficiency of its operation. The cost of the machine is ₹ 10 00 000 with a life period of 5 years. The machine can generate the benefit of ₹ 320 000 at the end of each year for first 4 years and ₹ 100 000 at the end of 5th year. Decide the Internal Rate of Return(IRR) of investing on the new machine by the Hit and Trial method. If the Company requires a minimum return of 11%, should it purchase the machine? [5]

5. (a) Establish the return to scale of the following situations and present them with proper figures: [5]

(i) Inputs Labour (L) and Capital (K) are increased by 1.5 times, as a result the total output increased from 200 units to 450 units.

(ii) In a production function $Q = f(L, K)$ where L is Labour and K is Capital, When the producer scale the inputs by t time ($t>0$) the output becomes $Q' = t.Q$

(b) (i) Given the Total Cost (TC) function of a company as $TC = 100+10Q+Q^2$. [5]

Find the Quantity (Q) when Average Cost (AC) and Marginal Cost (MC) are equal. Determine the value of AC and MC at this Quantity. Is the AC minimum at this Quantity? Why?

(ii) Prove that Average Variable Cost (AVC) is directly proportional to the reciprocal of Average Product (AP_L). Draw the AVC curve with the help of AP_L curve and write their relations.

6. (a) A dump truck is purchased for waste management by a local Municipal Body. The cost of the truck is \$110,000. The estimated salvage value at the end of the recovery period is \$8000. Prepare a schedule of annual depreciation and book value for the dump truck using the 200% Declining Balance method with a recovery period of five years. [5]

(b) A new bridge on river Valmiki collapsed due to heavy rain. For the benefit of the public the state government is planning to construct another bridge in its place with strong pile foundation and Reinforced Cement Concrete (RCC) structure. The estimated initial investment to construct the bridge is ₹20 00 00 000. The first period of the life of the bridge is 20 years after which it requires [5]

structural protection measures. This project requires an annual maintenance cost of ₹50,00 000. A Non Destructive Test (NDT) by the consultant at the end of 10th year needs an expense of ₹ 300 00 000. This bridge will provide benefit to the public in the form of faster transportation, good business, reducing fuel cost and saving lives and resources in emergencies. The annual benefit to the public is estimated to be ₹ 200 00 000. A scrap value of ₹100 00 000 is to be received from the bridge when structural protection measures will be given to extend the life period of the bridge. Calculate the Benefit-cost ratio of this project and comment on the implementation of the project. The rate of interest is 10% compounded annually (Use Annual Worth Method).

7. (a) A person is planning to start a small enterprise. This requires a fixed cost of ₹2 00 000. The variable cost ₹40 is to be incurred per unit of the product. The selling price of the product is ₹50 per unit. Since the break-even point is important for the managerial decision making, you work on the following points. [5]

   (i) - Finding Break-even quantity
   - Finding the influence of a 20% increase in the selling price on the break-even point.
   - Finding Break-even sales value using profit volume ratio (P/V ratio)

   (ii) Showing margin of safety when the actual sales is 25000 units in terms of a right diagram.

   (b) Inflation has become an everyday issue for the people now. Although causes of inflation are always open to discussions and arguments, on the basis of the sources it is classified into two broad categories viz Demand-pull inflation and Cost-push inflation. Considering that inflation is not only an economic phenomenon but also critical from policy perspectives, explain these two broad categories of inflation with suitable diagrams. [5]

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